

Module - 01

Introduction to Ethical Hacking

→ What is Ethical Hacking?

Ethical Hacking is authorized security testing to find vulnerabilities in systems, networks, & applications - before malicious hackers can exploit them.

→ Key Terms:

- Threat Actor: Who may attack you.
- Vulnerability: Weakness or Loop hole.
- Exploit: Code/method to abuse the weakness.
- Risk: Risk is probability of a threat exploiting a vulnerability.

→ Elements of Information Security (CIA Triad):

1. Confidentiality — Only authorized person can access.
2. Integrity — Data remain accurate & unaltered.
3. Availability — Data & Systems are Accessible.
4. Authenticity — Identity & data are genuine and verified.
5. Non-Repudiation — Actions or messages cannot be denied later.

→ Attacks = Motive (Goal) + Method (TTP) + Vulnerability (Weakness)

1. Motive — The Target System.
2. Method — TTP
3. Vulnerability — Weakness | Loop holes.

✚ Real-world ethical hackers sign NDA (Non-Disclosure Agreement) before testing and follow Rules of Engagement (ROE).

→ **Cyber Kill Chain:** Security Framework, process to compromise a system.

1. Reconnaissance - Gather data of target.
2. Weaponization - Creates Malicious Payload.
3. Delivery - Send Weaponized payload to target.
4. Exploitation - Payload Executes
5. Installation - Malware installs backdoors to maintain persistence.
6. C2 - System connects back to attacker's server for instruction.
7. Actions on objective - Attacker Completed the goal. [Data exfiltration, account takeover]

→ **MITRE ATT&CK:** MITRE is a Framework and Adversarial Tactic Technique & Common Knowledge it's a huge database of tactic and techniques that APT uses.

TTP —

- Tactics (Why) - Goal of the attacker
- Techniques (How) - Method used to achieve goal
- Procedures (Specific) - Exact tool or command used.

Module - 02

Footprinting and Reconnaissance

→ **Footprinting:** Footprinting is a process of gathering information about the target system, network or organization to identify potential vulnerabilities.

→ **Types of Reconnaissance:**

Passive Footprinting	Active Footprinting
Gathering information without direct interaction.	Gathering information with direct interaction.
Ex - Search Engines, Google Hacking Database (GHDB), Whois and Social Media.	Ex - DNS querying, Port Scanning.
Tools - Shodan, OSINT Framework, GDHDB	Tools - Nmap, Traceroute, Nslookup

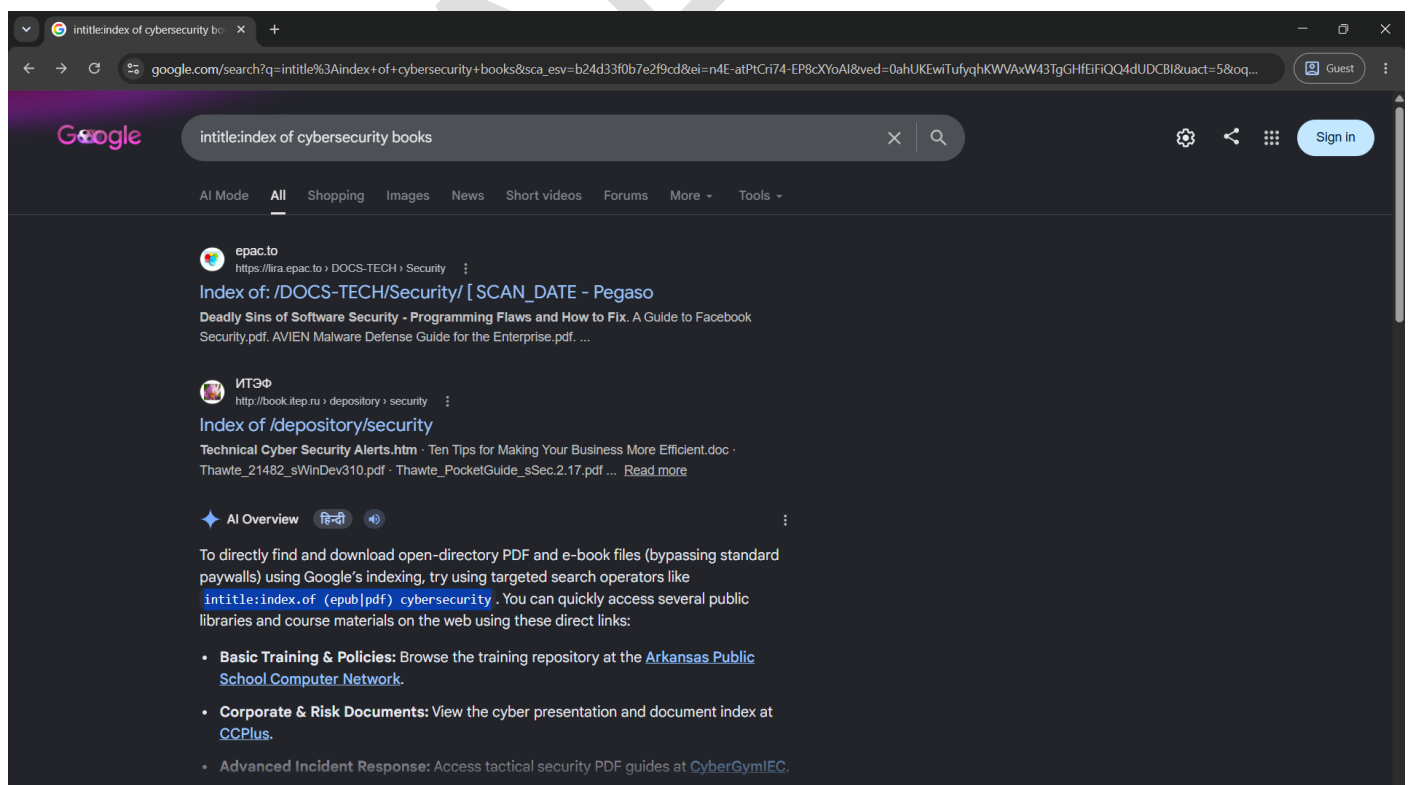
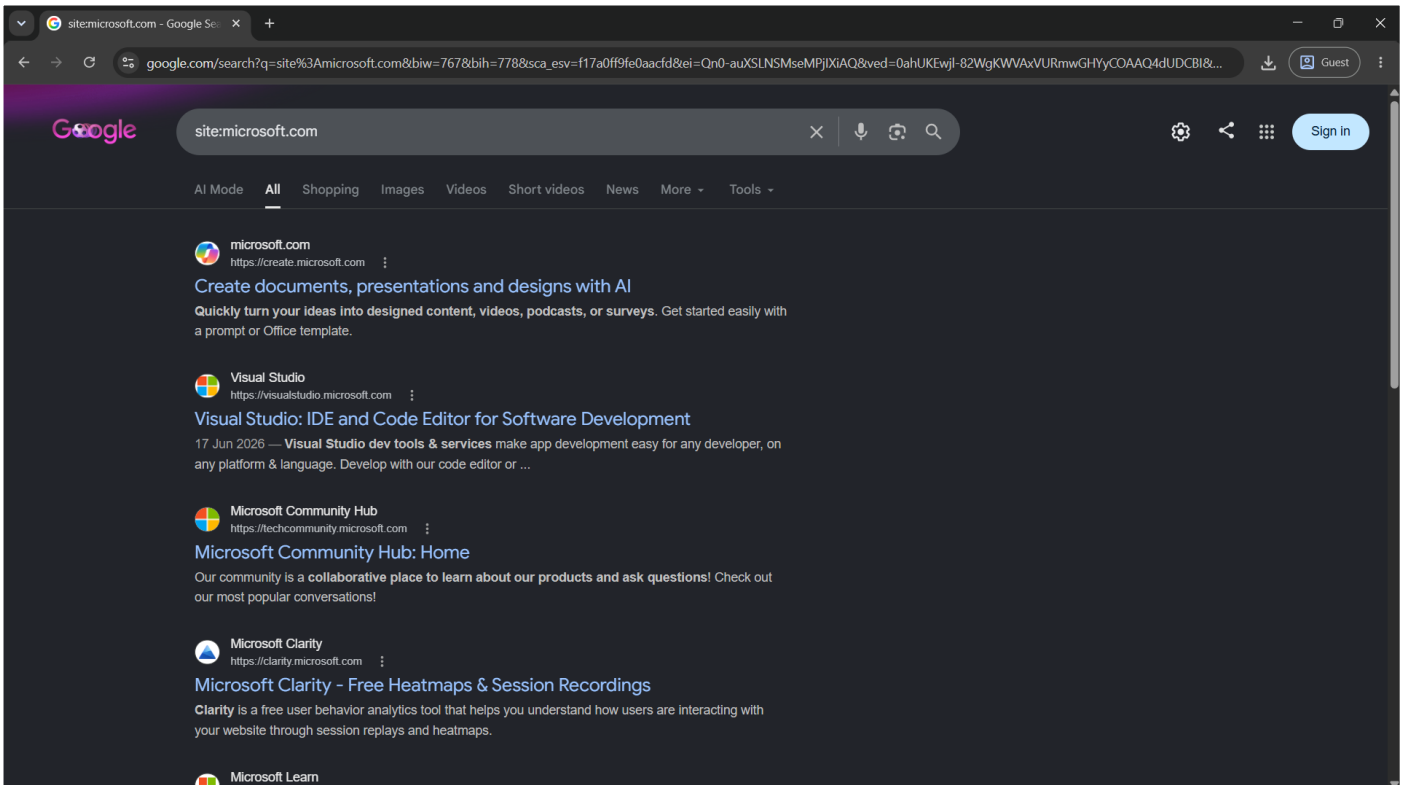
→ **Major Search Engines:** Google, DuckDuckGo, Bing, Yahoo, Ask.

- Attackers use search engines to extract information about a target or can used to find all other sources of publicly accessible information resources.
- Information that can be found: Domain Information, Subdomains, employee details, IP Address, Email IDs, DNS Records.

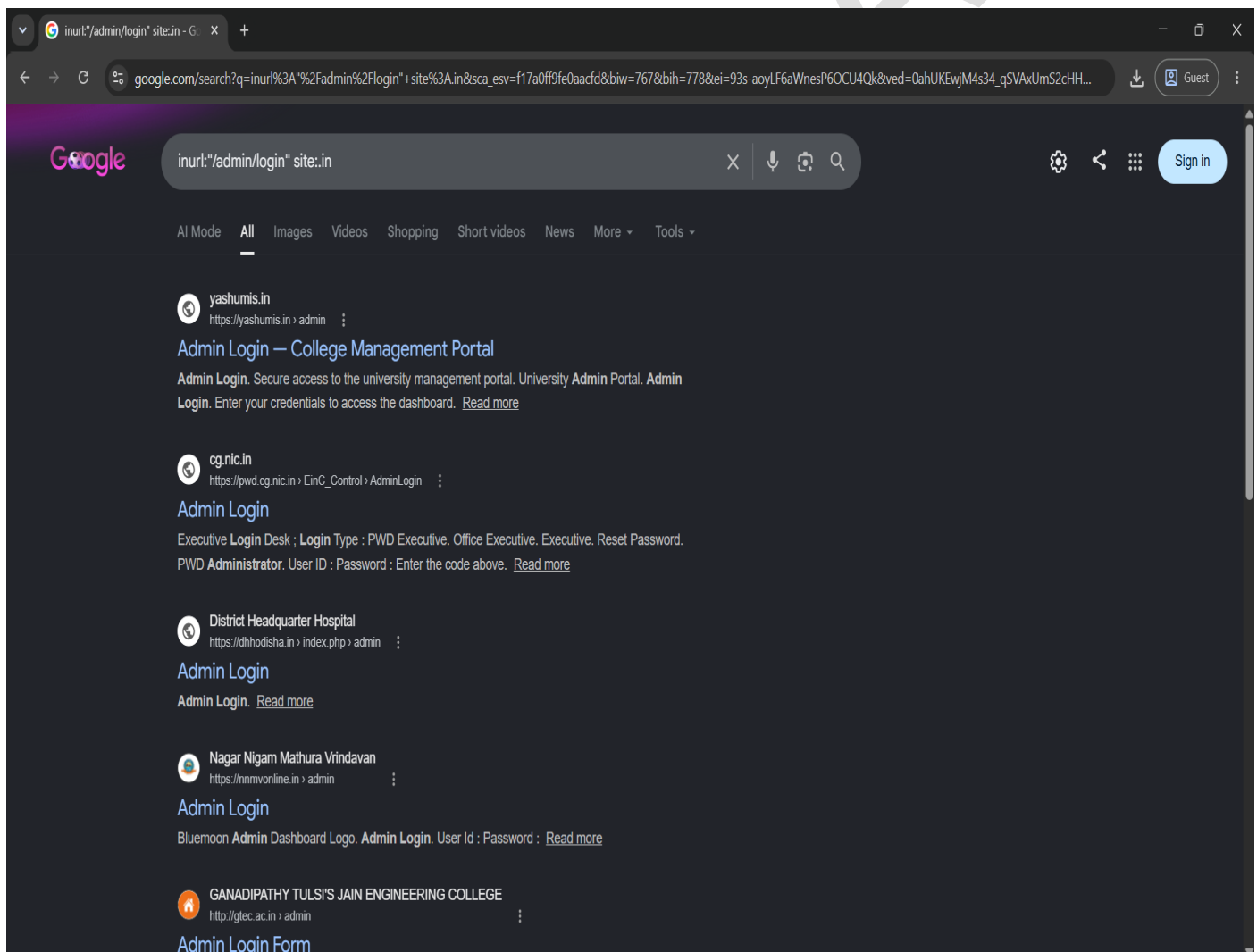
🌐 Search engines use automated software - crawlers, to continuously scan active websites & retrieve results in search engine index that further stored in massive database.

Google Hacking / Google Dorks

Google Hacking refers to advanced google search operators for creating complex search queries to extract sensitive or hidden information.

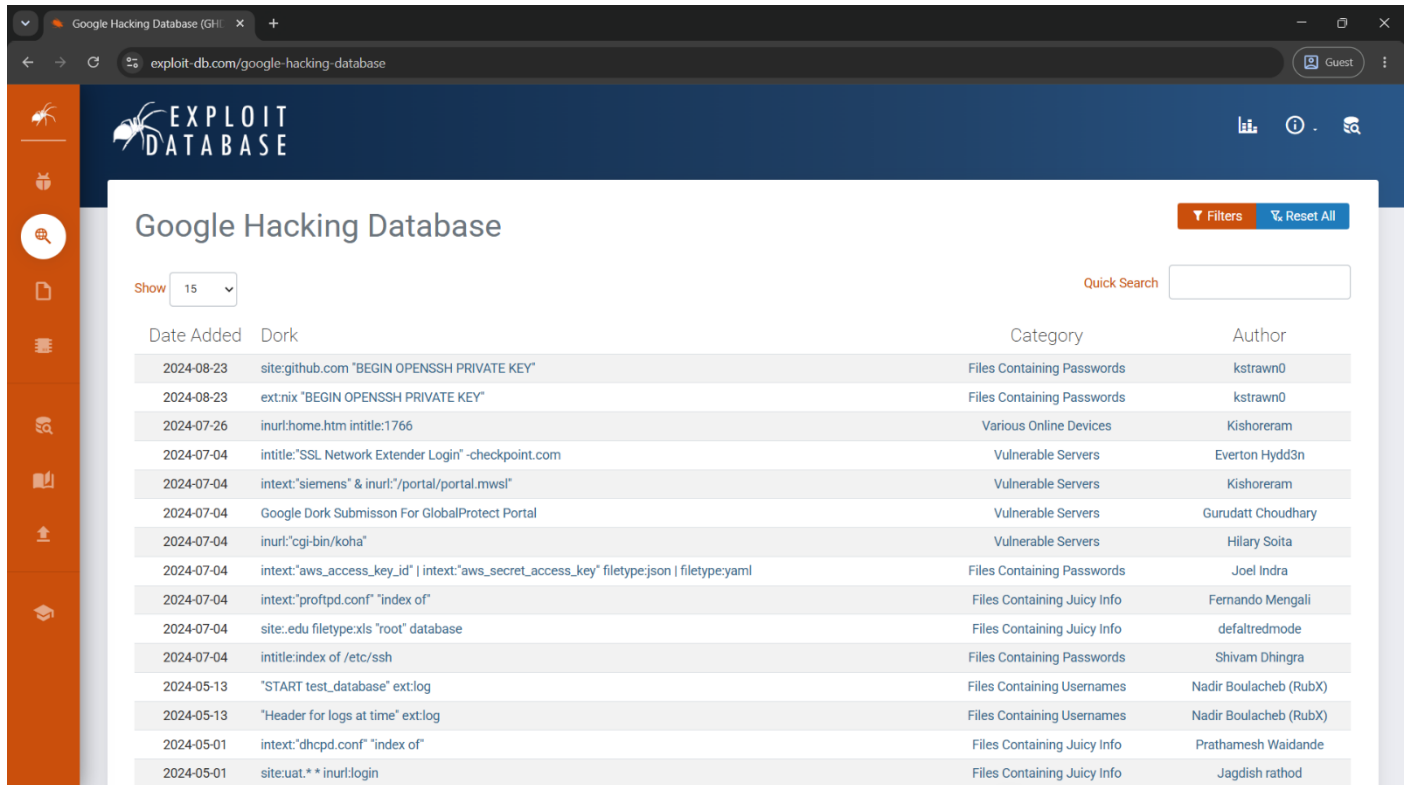


Search Operator	Purpose
[site:]	Restricts the result to specific domain.
[inurl:]	Restricts result to web pages containing specific keywords.
[cache:]	Displays web pages stored in Google cache.
[intitle:]	To find specific keywords in their title tags.
[info:]	Presents information that google has for that particular page.
[location:]	Finds information for a specific location.

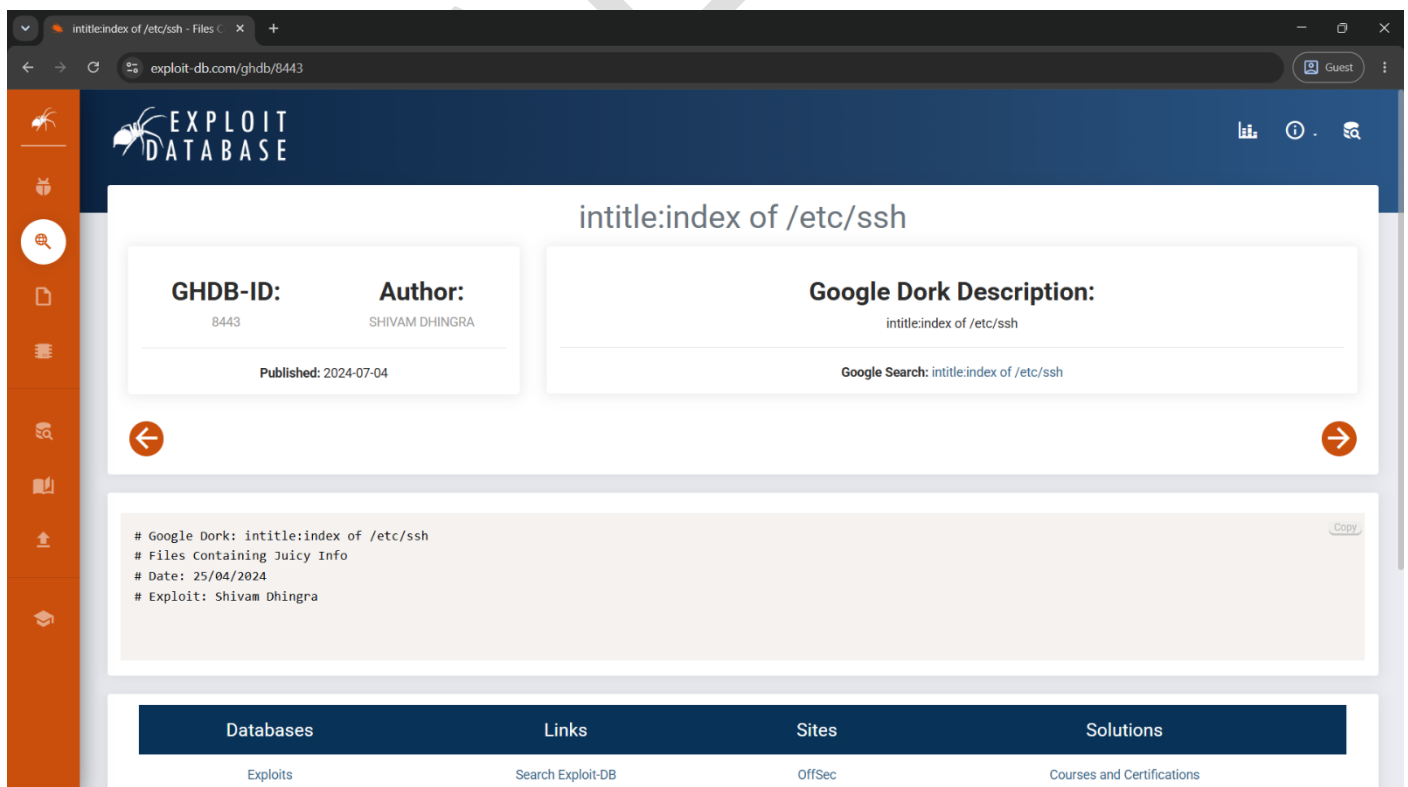


Google Hacking Database

The GHDB - Google Hacking Database is an authoritative source for querying the ever-widening reach of Google search engine.



Date Added	Dork	Category	Author
2024-08-23	site:github.com "BEGIN OPENSSSH PRIVATE KEY"	Files Containing Passwords	kstrawn0
2024-08-23	extrnix "BEGIN OPENSSSH PRIVATE KEY"	Files Containing Passwords	kstrawn0
2024-07-26	inurl:home.htm intitle:1766	Various Online Devices	Kishoreram
2024-07-04	intitle:"SSL Network Extender Login" -checkpoint.com	Vulnerable Servers	Everton Hydd3n
2024-07-04	intext:"siemens" & inurl:"/portal/portal.mwsl"	Vulnerable Servers	Kishoreram
2024-07-04	Google Dork Submission For GlobalProtect Portal	Vulnerable Servers	Gurudatt Choudhary
2024-07-04	inurl:"cgi-bin/koha"	Vulnerable Servers	Hilary Soita
2024-07-04	intext:"aws_access_key_id" intext:"aws_secret_access_key" filetype:json filetype:yaml	Files Containing Passwords	Joel Indra
2024-07-04	intext:"proftpd.conf" "index of"	Files Containing Juicy Info	Fernando Mengali
2024-07-04	site:.edu filetype:xls "root" database	Files Containing Juicy Info	defaltredmode
2024-07-04	intitle:index of /etc/ssh	Files Containing Passwords	Shivam Dhingra
2024-05-13	"START test_database" ext:log	Files Containing Usernames	Nadir Boulacheb (RubX)
2024-05-13	"Header for logs at time" ext:log	Files Containing Usernames	Nadir Boulacheb (RubX)
2024-05-01	intext:"dhcpd.conf" "index of"	Files Containing Juicy Info	Prathamesh Waidande
2024-05-01	site:uat.* * inurl:login	Files Containing Juicy Info	Jagdish rathod



intitle:index of /etc/ssh

GHDB-ID: 8443
Author: SHIVAM DHINGRA
Published: 2024-07-04

Google Dork Description: intitle:index of /etc/ssh
Google Search: intitle:index of /etc/ssh

Google Dork: intitle:index of /etc/ssh
Files Containing Juicy Info
Date: 25/04/2024
Exploit: Shivam Dhingra

Databases **Links** **Sites** **Solutions**

Exploits Search Exploit-DB OffSec Courses and Certifications

The exploit Database is a Common Vulnerabilities and Exposures (CVE) compliant archive of public exploits and corresponding vulnerable software.

Whois

Whois: Whois is a query and response protocol used of querying databases that store the registered user's details.



DomainTools

PROFILE ▾

CONNECT ▾

MONITOR ▾

SUPPORT ▾

Whois Lookup



[Home](#) > [Whois Lookup](#) > [Asdacademy.in](#)

Notice: Possible deprecatio

Whois Record for Asdacademy.in

— Domain Profile

Registrar	HOSTINGER operations, UAB IANA ID: 1636 URL: http://www.hostinger.com Whois Server: whois.hostinger.com domains@hostinger.com (p) +3.7068424669
Registrar Status	clientTransferProhibited
Dates	2,193 days old Created on 2020-06-26 Expires on 2027-06-26 Updated on 2026-06-05
Name Servers	NS1.DNS-PARKING.COM (has 9,953,350 domains) NS2.DNS-PARKING.COM (has 9,953,350 domains)
IP Address	145.79.209.146 - 151 other sites hosted on this server
IP Location	- Maharashtra - Mumbai - Hostinger International Limited
ASN	AS47583 AS-HOSTINGER Hostinger International Limited, CY (registered Apr 04, 2011)
IP History	8 changes on 8 unique IP addresses over 4 years
Hosting History	7 changes on 4 unique name servers over 6 years

Whois Record (last updated on 2026-06-29)

Domain Name: asdacademy.in
Registry Domain ID: D0A47DE316580450691E6E871F07361D0-IN
Registrar WHOIS Server: whois.hostinger.com

- 🚦 Regional Internet Registries (RIRs) regional IP Manager, maintain Whois databases, which contain personal information of domain owner.
- 🚦 5 RIRs - ARIN, RIPE NCC, APNIC, LACNIC and AFRINIC.

Whois query returns the following information:

Field	Value	Description
Domain Name	asdacademy.in	Registered website address.
Registrar	HOSTINGER operations	Company that registered the domain.
Whois Server	Whois.hostinger.com	Server storing domain registration records.
IANA ID	1636	Unique identifier of registrar.
Dates	Created = 26 June 2020 Expires = 26 June 2027 Updated = 5 June 2026	Created, Expiry and updated details.
Name Servers	NS1.DNS-PARKING.COM NS2.DNS-PARKING.COM	Servers directing the domain to its hosting.
IP Address	145.79.209.146	IP Address of hosting server.
IP Location	Mumbai, Maharashtra	Geographic Location
ASN	AS47583	Autonomous System Number of Network Provider.

- ✚ IANA - Internet Assigned Numbers Authority that distributes large blocks of IP Addresses to Five RIRs, IANA is managed by ICANN.
- ✚ ICANN - Internet Corporation for assigned Names and Numbers is an organization.
- ✚ ASN - Autonomous System Number, Unique identifier for network that owns or routes the domain's IP address.

Netcraft

Netcraft: It is utilized as an OSINT tool to map the target details.

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Background

Site title	Your request has been blocked. This could be due to several reasons.	Date first seen	August 1995
Site rank	22294	Primary language	English
Description	Not Present		

Network

Site	http://microsoft.com	Domain	microsoft.com
Netblock Owner	Microsoft Corporation	Nameserver	ns1-39.azure-dns.com
Hosting company	Microsoft Corporation	Domain registrar	markmonitor.com
Hosting country	US	Nameserver organisation	whois.markmonitor.com
IPv4 address	150.171.109.245 (VirusTotal)	Organisation	Microsoft Corporation, One Microsoft Way., Redmond, 98052, United States
IPv4 autonomous systems	AS8075	DNS admin	azuredns-hostmaster@microsoft.com
IPv6 address	2603:1061:14:f0:0:0:0:1	Top Level Domain	Commercial entities (.com)
IPv6 autonomous systems	AS8075	DNS Security Extensions	Enabled
Reverse DNS	Unknown		

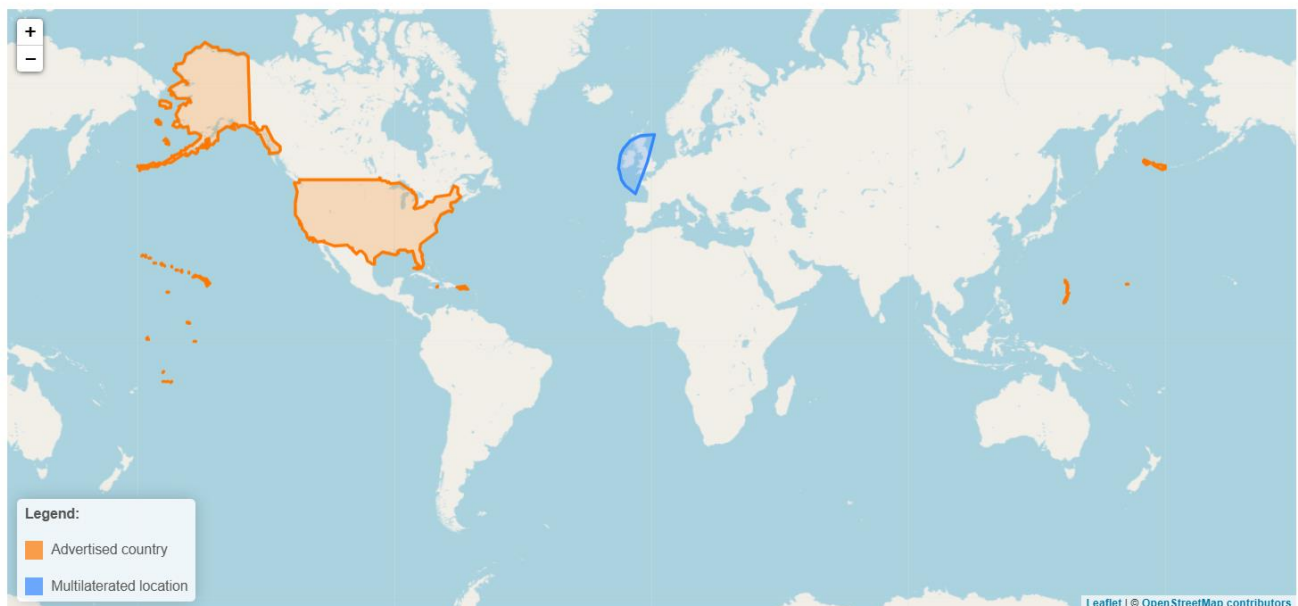
IP delegation

IPv4 address (150.171.109.245)

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IP Geolocation

We use multilateration to independently determine the location of a server. [Read more.](#)



Analyses: -

Field	Value	Description
Website	http://microsoft.com	Target website URL.
Netblock Owner	Microsoft Corporation	Organization owning the IP address block.
Hosting Company	Microsoft Corporation	Company hosting the website.
IPv4 Address	150.171.109.245	Public IPv4 address of the web server.
IPv6 Address	2603:1061:14:f0::1	Public IPv6 address of the web server.
ASN	AS8075	Autonomous System Number identifying Microsoft's network.
Domain	Microsoft.com	Registered domain name.
Domain Registrar	MarkMonitor	Company managing the domain registration.
Top-Level Domain	.com	Commercial top-level domain.
Name Server	ns1-39.azure-dns.com	DNS server responsible for resolving the domain.
Name Server Organization	whois.markmonitor.com	WHOIS service associated with the registrar.
DNS Administrator	azuredns-hostmaster@microsoft.com	Administrative contact for DNS management.
SPF	Configured	Specifies authorized mail servers that can send emails for the domain.
DMARC	p=reject	Rejects emails that fail authentication checks.

Shodan

Shodan: It is a search engine for internet-connected devices, allows researchers to find exposed system without directly touching the target.

Target: Microsoft.com

The screenshot shows the Shodan search results page for the query 'microsoft.com'. The page has a dark theme. At the top, there's a navigation bar with 'SHODAN' and links for 'Explore', 'Downloads', 'Pricing', and 'Account'. Below the navigation bar, the search results are displayed. On the left, there's a 'TOTAL RESULTS' section showing '36,686' results. Below that, a 'TOP COUNTRIES' section shows a world map and a list of countries: United States (9,577), Germany (7,235), Netherlands (2,928), China (2,162), and Italy (1,702). Further down, a 'TOP PORTS' section shows a list of ports: 443 (22,960), 80 (8,663), 8443 (1,972), and 4443 (1,436). On the right, there's a 'Document Moved' section with details about a document from 'United States, New York City'. Below that, there's an 'HTTP Status 404 Not Found' section with details about a status code. At the bottom, there's an 'SSL Certificate' section with details about a certificate issued by 'DigiCert Inc'.

The screenshot shows the Shodan host details page for the IP address '52.105.237.39'. The page has a dark theme. At the top, there's a navigation bar with '52.105.237.39' and links for 'Regular View', 'Raw Data', 'Timeline', and 'Whois'. Below the navigation bar, there's a 'General Information' section with details about the host: Country (India), City (Pune), Organization (Microsoft Corporation), ISP (Microsoft Corporation), ASN (AS8075), and Operating System (Windows). To the right, there's an 'Open Ports' section showing a list of open ports: 80 and 443. Below that, there's a 'Web Technologies' section with details about the web technologies used: CMS (Microsoft SharePoint 16.0.0.27410), Web Frameworks (Microsoft ASP.NET), Operating Systems (Windows Server), and Web Servers (IIS 10.0). At the bottom, there's a 'Vulnerabilities' section showing a list of vulnerabilities: 2, 114, 56, 2, and 0.

Analysis:

Field	Value	Description
Target IP Address	52.105.237.39	Public IP address identified during reconnaissance.
Organization/Owner	Microsoft Corporation	The IP address is registered to Microsoft.
ASN	AS8075	Microsoft's Autonomous System Number used for Internet routing and network ownership identification.
Geographic Location	Pune, India	Physical Location
Operating System	Windows Server	Operating System hosting on Web services.
Web Server	Microsoft IIS 10.0	Internet Information Services (IIS) version 10.0 is used to serve web applications.
Port - 80	HTTP	HTTP service
Port - 443	HTTPS	HTTPS service

Sublist3r

Sublist3r: Python-based reconnaissance tool designed to enumerate subdomains of websites using Open-Source Intelligence (OSINT).

```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home X kali-linux-2026.1-vmware-a... X
1 2 3 4
Minimize all open windows and show the desktop
Session Actions Edit View Help

(root@kali)-[~]
# sublist3r -d google.com

# Coded By Ahmed Aboul-Ela - @aboul3la

[-] Enumerating subdomains now for google.com
[-] Searching now in Baidu..
[-] Searching now in Yahoo..
[-] Searching now in Google..
[-] Searching now in Bing..
[-] Searching now in Ask..
[-] Searching now in Netcraft..
[-] Searching now in DNSDumpster..
[-] Searching now in Virustotal..
[-] Searching now in ThreatCrowd..
[-] Searching now in SSL Certificates..
[-] Searching now in PassiveDNS..
[!] Error: Virustotal probably now is blocking our requests
[!] DNSDumpster module failed: Could not find CSRF token on DNSDumpster page
[-] Total Unique Subdomains Found: 180
www.google.com
accounts.google.com
admanager.google.com
admin.google.com
admob.google.com
ads.google.com
adsense.google.com
adssettings.google.com
adstransparency.google.com
aistudio.google.com
analytics.google.com
apis.google.com
uc.appengine.google.com
apps.google.com
artsandculture.google.com
assignments.google.com
assistant.google.com
books.google.com
bughunters.google.com
business.google.com
calendar.google.com
chat.google.com
chrome.google.com
chromewebstore.google.com

kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home X kali-linux-2026.1-vmware-a... X
1 2 3 4
Session Actions Edit View Help

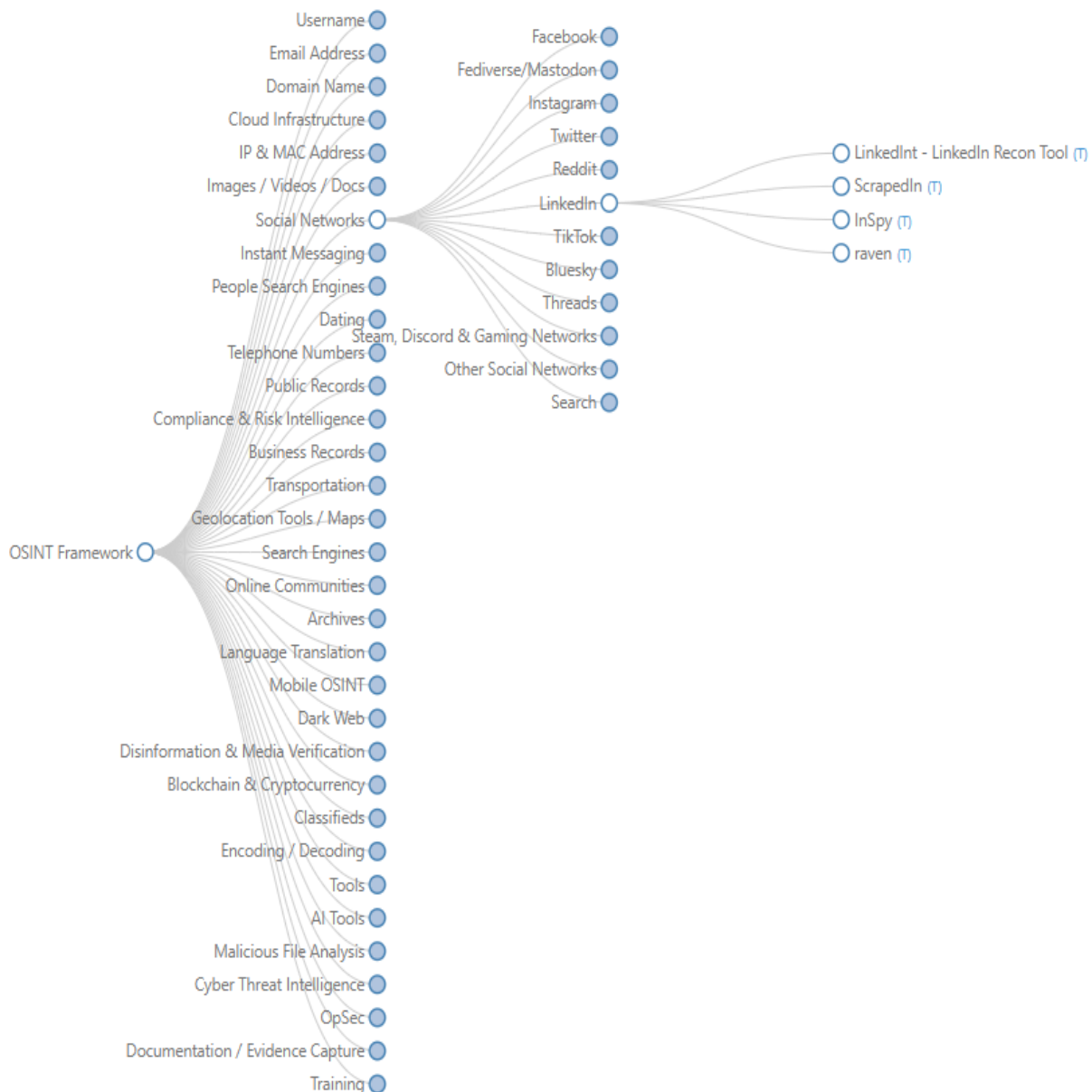
patents.google.com
pay.google.com
payments.google.com
photos.google.com
play.google.com
policies.google.com
postmaster.google.com
profile.google.com
programmablesearchengine.google.com
publishercenter.google.com
recorder.google.com
remotedesktop.google.com
reportcontent.google.com
research.google.com
colab.research.google.com
datasetsearch.research.google.com
safebrowsing.google.com
santatracker.google.com
scholar.google.com
script.google.com
search.google.com
status.search.google.com
services.google.com
sites.google.com
startup.google.com
store.google.com
support.google.com
tagassistant.google.com
tagmanager.google.com
takeout.google.com
tasks.google.com
toolbox.google.com
tools.google.com
translate.google.com
transparencyreport.google.com
trends.google.com
chat.usercontent.google.com
contribution.usercontent.google.com
drive.usercontent.google.com
photos.fife.usercontent.google.com
vault.google.com
voice.google.com
wallet.google.com
workspace.google.com
access.workspace.google.com
flows.workspace.google.com
knowledge.workspace.google.com
studio.workspace.google.com

(root@kali)-[~]
```

OSINT Framework

OSINT Framework

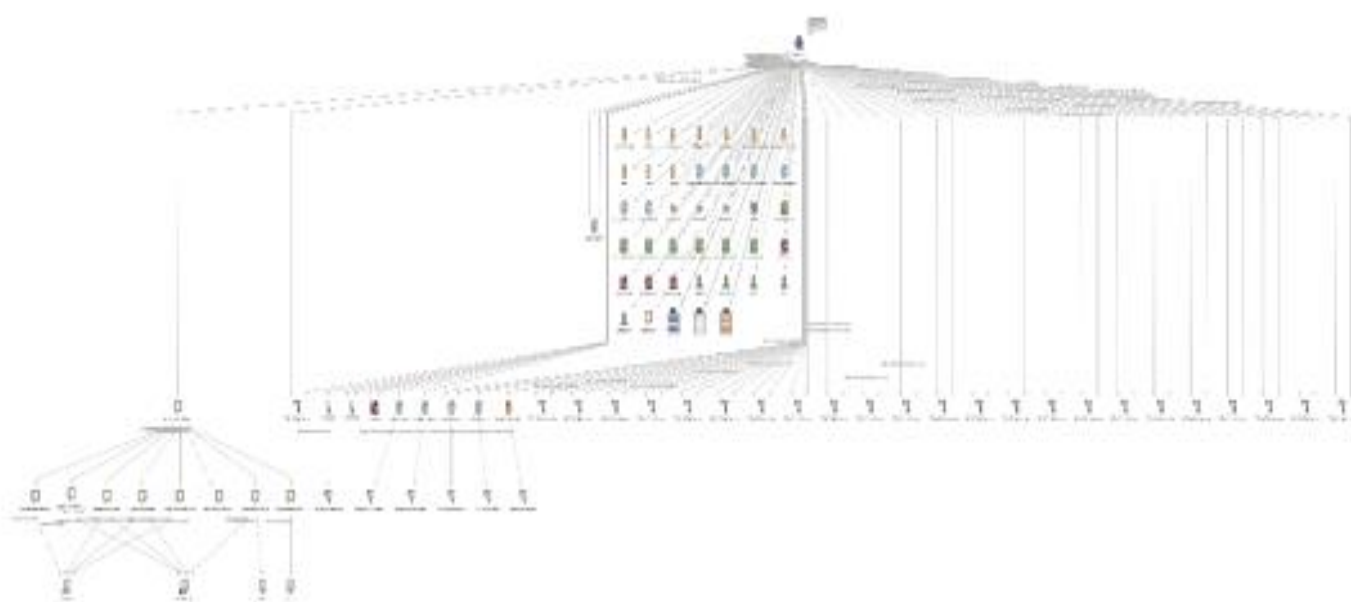
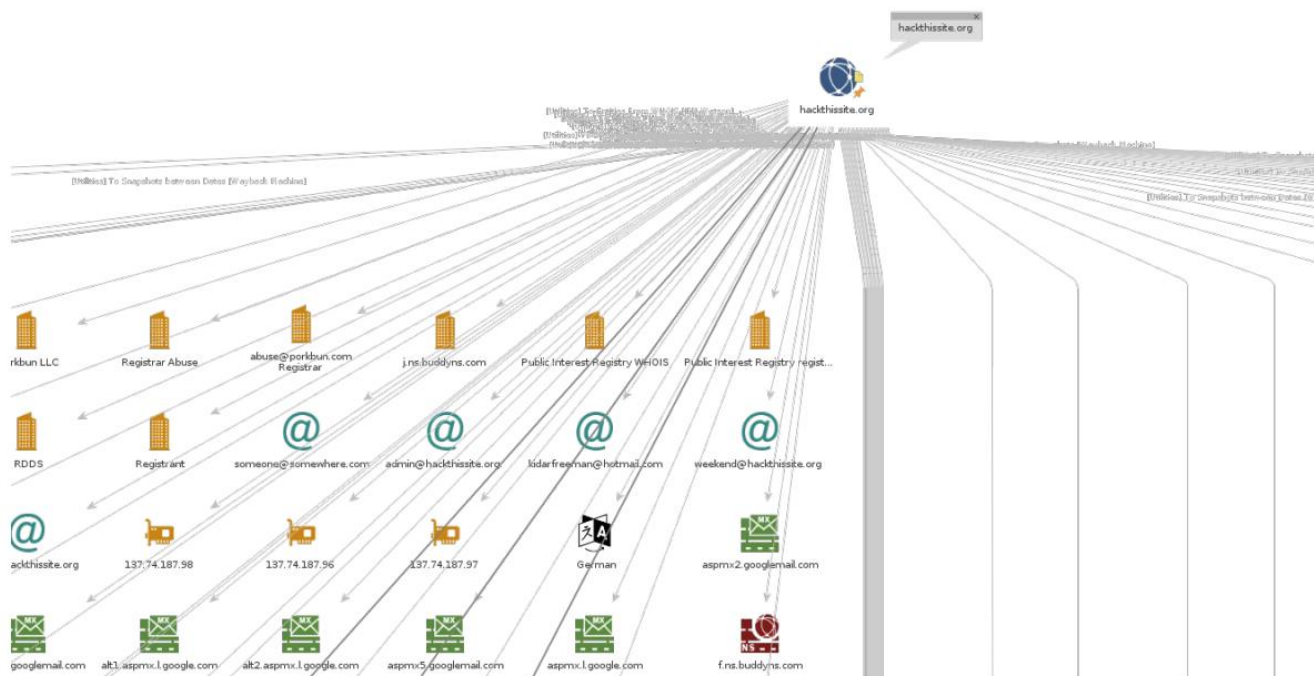
Search tools...



🚀 OSINT Framework is a massive, structured collection of Open-source Intelligence tools and resources.

Maltego

Maltego: It is cyber investigation and OSINT platform used to collect, correlate, and visually analyze data.



Recon-ng

Recon-ng: Web Reconnaissance framework with independent modules, database interaction, CLI based and open-source web-based framework.

Commands used:

- marketplace install all — To install all modules.
- modules search — To load all modules.
- modules load <netcraft> — To use the module.
 - >options set SOURCE google.com
 - >info
 - >run
- workspaces create my_workspace — To create own workspace.

```
[recon-ng][default] > help
```

```
Commands (type [help|?] <topic>):
```

back	System	Exits the current context
dashboard		Displays a summary of activity
db		Interfaces with the workspace's database
exit		Exits the framework
help		Displays this menu
index		Creates a module index (dev only)
keys		Manages third party resource credentials
marketplace		Interfaces with the module marketplace
modules		Interfaces with installed modules
options		Manages the current context options
pdb		Starts a Python Debugger session (dev only)
script		Records and executes command scripts
shell		Executes shell commands
show		Shows various framework items
snapshots		Manages workspace snapshots
spool		Spools output to a file
workspaces		Manages workspaces

```
[recon-ng][default] > workspaces
```

```
Manages workspaces
```

```
Usage: workspaces <create|list|load|remove> [ ... ]
```

```
[recon-ng][default] > workspaces list
```

+	-----+	
	Workspaces	Modified
+	-----+	
	default	2026-06-29 06:26:30
	my_workspace	2026-06-29 06:29:35
+	-----+	

```
[recon-ng][default] > workspaces load my workspace
```



```
Home X kali-linux-2026.1-vmware-a... X
1 2 3 4
root@kali: ~
Session Actions Edit View Help
[recon-ng][default][netcraft] > source google.com
[!] Invalid command: source google.com.
[recon-ng][default][netcraft] > options set SOURCE google.com
SOURCE => google.com
[recon-ng][default][netcraft] > info
    Name: Netcraft Hostname Enumerator
    Author: thrapt (thrapt@gmail.com)
    Version: 1.1
Description:
    Harvests hosts from Netcraft.com. Updates the 'hosts' table with the results.
Options:
    Name      Current Value  Required  Description
    SOURCE    google.com      yes       source of input (see 'info' for details)
Source Options:
    default      SELECT DISTINCT domain FROM domains WHERE domain IS NOT NULL
    <string>      string representing a single input
    <path>        path to a file containing a list of inputs
    query <sql>  database query returning one column of inputs
[recon-ng][default][netcraft] > run
-----
GOOGLE.COM
[*] URL: http://searchdns.netcraft.com/?restriction=site%2Bends%2Bwith&host=google.com
[*] Country: None
[*] Host: www.google.com
[*] Ip_Address: None
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
-----
[*] Country: None
[*] Host: myaccount.google.com
[*] Ip_Address: None
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
-----
[*] Country: None
[*] Host: accounts.google.com
[*] Ip_Address: None
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.